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END TERM EXAMINATION
SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: BS-106

Subject: Applied Physics-II

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q. No.	Question	Marks
Q1	Answer the following questions:	(3×5=15)
(a)	An eigen function of an operator d^2/dx^2 is $\Psi = e^{ax}$. Find the corresponding Eigen value.	(3)
(b)	Show that Wien's law is a special case of Planck's radiation law.	(3)
(c)	The lattice constant of a cubic lattice is a. Calculate the spacing between (011), (101), (112), (111), and (100) planes.	(3)
(d)	Explain why a semiconductor acts as an insulator at 0 K.	(3)
(e)	The wave function of a particle is $\Psi = A \cos^2 x$ for the interval $-\pi/2 \leq x \leq \pi/2$. Find the value of A.	(3)
UNIT - I		
Q2 (a)	What do you mean by particle in a box? Show that the energy of an electron in the box varies as the square of the natural numbers.	(3 + 7 = 10)
(b)	Prove quantum mechanically that a particle will not exist in a box if its energy is zero.	(3)
(c)	Differentiate between Ψ and $ \Psi ^2$.	(2)
OR		
Q3 (a)	Distinguish between phase velocity and group velocity. Show that the de-Broglie wave group associated with a moving particle travels with the same velocity as the particle.	(3 + 7 = 10)
(b)	What is the physical significance of wave function?	(3)
(c)	The particle trapped in a one-dimensional box of length L is described by a wave function $\Psi = x$. Normalise the wave function between a and b.	(2)

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Q. No.	Question	Marks
UNIT - II		
Q4 (a)	State Planck's formula for Black body radiation and derive it from BE statistics.	(8)
(b)	If the Sun has a surface temperature of 5700 K, what is the wavelength of maximum intensity of solar radiation?	(4)
(c)	Define Stefan's law.	(3)
OR		
Q5 (a)	An electron gas obeys Maxwell-Boltzmann statistics. Calculate the average thermal energy (in eV) of an electron in the system at 300 K.	(4)
(b)	Distinguish between Bosons and Fermions.	(3)
(c)	What is the relative population of the first two single-particle energy levels of a system of distinguishable particles if the energies of the levels are $\epsilon_0 = 0$ and $\epsilon_1 = kT$?	(4)
(d)	Discuss the Fermi-Dirac distribution with the help of the distribution function; explain the concepts of Fermi level and Fermi energy.	(4)
UNIT - III		
Q6 (a)	Why do we use X-rays to study crystal structure?	(3)
(b)	State Bragg's law. How is it applied?	(3)
(c)	Write short notes on: (i) Point defects (ii) Frenkel defects (iii) Schottky defects	(9)
OR		
Q7 (a)	Define the following: (i) Unit Cell (ii) Space Lattice (iii) Coordination number (iv) Miller indices	(2+2+2+2=8)
(b)	Deduce Miller indices for the plane having intercepts a, b, and c at -2, ∞ , -2. Also draw the plane.	(4)
(c)	X-rays of wavelength $2 \times 10^{-11} \text{ m}$ suffer first-order reflection from the (111) crystal plane at an angle of 45° . What is the interatomic spacing of the crystal?	(3)
UNIT - IV		
Q8 (a)	Distinguish between intrinsic and extrinsic semiconductors.	(4)
(b)	Explain Kronig-Penney model for the motion of an electron in a periodic potential.	(7)
(c)	Differentiate between conductor, insulator, and semiconductor using energy-band diagrams.	(4)
OR		
Q9 (a)	Show that the Fermi energy lies midway between the conduction band and valence band for an intrinsic semiconductor.	(6)
(b)	Write short notes on: (i) Zener diode (ii) PN junction diode (iii) Photodiode	(3+3+3=9)

**** End of Question Paper ****